semanticfa: An R Package for Response-Free Semantic Factor Analysis of Psychometric Scales

Devon Yanitski[1] and Chris Westbury[1]

[1]Department of Psychology, University of Alberta, Edmonton, Alberta, Canada

Author Note

Correspondence concerning this article should be addressed to Devon Yanitski, Department of Psychology, University of Alberta, Edmonton, Alberta, Canada. E-mail: dyanitsk@ualberta.ca

Abstract

The items of a psychometric scale constitute a proposal, written in ordinary language, about the structure of a construct—and language-model embeddings make that proposal measurable before a single respondent is recruited. We introduce `semanticfa`, an R package for semantic factor analysis (SFA): exploratory factor analysis of the similarity structure of language-model embeddings of scale items, requiring no response data at any stage. The package implements the complete workflow: similarity construction under keyed, keying-free, and natural-language-inference encodings; factor retention by multi-criterion consensus; extraction with Monte Carlo–calibrated diagnostics; interpretation by anchoring, semantic projection, congruence, and jingle–jangle screening; and response-free refinement (redundancy detection, short forms, candidate-item vetting). A fully reproducible demonstration exercises every exported function on the 50-item International Personality Item Pool Big-Five Factor Markers, embedding the items with Qwen3-Embedding-8B and validating the semantic structure against a conventional factor analysis of 874,434 respondents. The retention consensus indicated five factors, and the semantic solution recovered the Big Five domains, with congruence to the human factors graded by domain (Tucker's ϕ from .92 for Openness to .52 for Agreeableness) and semantic similarities correlating $r = .46$ with response correlations across all 1,225 item pairs. Sign-flipping reverse-keyed embeddings produced anti-topic vectors that aligned worst with keyed human data, and a semantically central 25-item short form recovered the theoretical structure better than the full inventory. We frame the agreement of the two structures—semantic–behavioral coherence—as an estimable relation in a construct's nomological network: a complement to, not a replacement for, response-based factor analysis. The package is available on CRAN.

Keywords: semantic factor analysis, language models, embeddings, construct validity, scale development, R package

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Every factor analysis of a rating scale begins with items written in ordinary language, and we take that wording seriously as data. A scale's items constitute a *semantic proposal*—a deliberate attempt, in language, to sample the content that constitutes a construct—and modern language-model embeddings make that proposal measurable before any respondent answers, mapping each item to a point in a vector space whose geometry encodes its meaning. We introduce `semanticfa` (Yanitski & Westbury, 2026), an R package for *semantic factor analysis* (SFA): exploratory factor analysis of the similarity structure of language-model embeddings of scale items, requiring no human response data at any stage.

The motivating distinction is between two relational structures over the same items. Classical factor analysis recovers a scale's *behavioral* structure, the latent dimensions of person-by-item response covariance; SFA recovers its *semantic* structure, the geometry of item meaning. These are structurally parallel analyses on different data sources, and their agreement, *semantic–behavioral coherence*, is an estimable relation in a construct's nomological network: convergence is intranetwork validity evidence, while divergence is itself informative, locating where the linguistic and behavioral renderings of a construct come apart. Throughout, SFA *complements*, never replaces, response-based factor analysis. We develop this in three movements—the theory of analyzing a scale's language, a review of language models in psychometrics, and the positioning of `semanticfa` and its demonstration.

Meaning as Use, Distributional Semantics, and Embeddings

A theory of word meaning is the natural starting point for analyzing a scale's language. Wittgenstein (1953, §43) holds, for "a large class of cases," that "the meaning of a word is its use in the language"; and a category like *games* shares no single essence, only "a complicated network of similarities"—*family resemblance* (Wittgenstein, 1953, §§66–67).

A construct, likewise, can cohere through graded, overlapping resemblances rather than a hidden shared property—the relational picture distributional similarity makes measurable.

Meaning-as-use supplies no method; *distributional* theory narrows it to a claim about co-occurrence. Joos (1950) defined a unit's meaning as the conditional probabilities of its occurrence with all others, and Harris (1954) the distribution of an element as the sum of its environments— "difference of meaning correlates with difference of distribution." Modern embeddings make this computable: Mikolov et al. (2013) learn word vectors whose offsets encode relations as arithmetic ($\text{king} - \text{man} + \text{woman} \approx \text{queen}$), and Pennington et al. (2014) factorize global word–word co-occurrence directly. Such spaces implement a theory of semantic memory, predicting human similarity at near-interrater levels (categorization, synonymy, and priming more loosely), with interpretable structure recoverable post hoc (Günther et al., 2019); a complementary prediction-based line matched human norms and released precomputed spaces behind a query interface (Mandera et al., 2017).

Scale items are sentences. The same intuition was scaled by the *transformer* (Vaswani et al., 2017), whose self-attention makes each token a function of its context, and by BERT's deep bidirectional *contextual* embeddings (Devlin et al., 2019). Because averaged BERT tokens compare worse than averaged static vectors, Reimers and Gurevych (2019) fine-tune a siamese Sentence-BERT placing similar sentences close, relatedness read off as cosine similarity. This capability—whole items in a metric space where proximity tracks meaning, alongside validated lexical co-occurrence spaces (Westbury, 2025)—is what SFA uses.

Constructs as Relations and Items as a Semantic Proposal

This relational picture of meaning prefigures one of constructs. Just as a word's meaning is fixed by use and a category coheres through family resemblance (Wittgenstein, 1953, §43, §§66–67), a construct is defined not as a hidden attribute but implicitly by its place in a *nomological network* of lawful relations to behavior, observation, theory, and other constructs, so two operations "measure the same thing" through an *intranetwork* identification rather than a direct comparison (Cronbach & Meehl, 1955). The relational conception thus recurs on both sides—distributional theory defines word meaning by co-occurrence, nomological theory construct meaning by lawful relation—and scales meet at the intersection.

Modern validity theory reinforces this: the score is one indicator whose convergent and discriminant relations substantiate meaning, with a *content* aspect (item relevance) and a *structural* aspect (scoring mirroring the domain) (Cronbach & Meehl, 1955; Messick, 1995). Items thus form an operationalized semantic proposal open to underrepresentation and irrelevant variance (Messick, 1995). Whether item meaning should mirror item covariance depends on the model: a *reflective* latent variable causes intercorrelated, internally consistent indicators, making factor analysis apt, whereas *formative* indicators jointly constitute the construct and need not correlate (Bollen & Lennox, 1991). SFA presupposes the reflective case dominating personality and attitude scales, where low coherence for formative sets is anticipated, not diagnostic (Bollen & Lennox, 1991). Within it, items deliberately sample content: a pool must be overinclusive of Loevinger's "all possible contents which might comprise the putative trait," and phrasing shapes measurement, since "almost any negative mood term ... virtually guarantees a substantial neuroticism" component (Clark & Watson, 2019).

This does not license reading structure off language alone. Borsboom et al. (2004) argue validity is ontological and causal: a test is valid for an attribute only if the attribute exists and causally produces the outcomes, so the nomological match of Cronbach and Meehl (1955) has a "signaling function," not a constitutive one. On this realist caveat, semantic structure is evidence about a construct's nomological network, not proof the attribute exists; the analysis is not circular, since both theories locate meaning relationally (Borsboom et al., 2004).

Two Structures Over the Same Items

A scale's items support two relational structures. The *behavioral* one has a long lineage. Galton (1884) proposed measuring character by tabulating "carefully recorded acts," and counted some thousand character words, each with "a separate shade of meaning, while each shares a large part of its meaning with some of the rest." Spearman (1904) made it a method: correlated series "may be forthwith assumed to indicate and measure something common to both." Pursuing Galton's second reading, Goldberg (1990) showed factor analyses of trait-adjective pools recover the Big Five across raters and procedures. SFA pursues the first reading, from item-meaning geometry; that the two often coincide is the coherence relation we estimate.

The behavioral machinery transfers. Dimensionality long rested on the latent-root-one rule, Guttman's weakest lower bound on the correlation matrix's rank (Kaiser, 1960), with interpretation by rotation invariant to the items sampled (Kaiser, 1958); because that rule capitalizes on chance, parallel analysis retains only eigenvalues exceeding those of random data (Horn, 1965). We adopt parallel analysis for SFA because it transfers unchanged from a respondent correlation matrix to an embedding similarity matrix (Horn, 1965). Confirmatory factor analysis, fixing loadings and testing whether a structure reproduces covariances (Jöreskog, 1969), furnishes the coherence template: a pattern from one source becomes a hypothesis another can corroborate, divergence localizing which relations fail. Reading meaning off a solution is an old aspiration: Lara et al. (1992) called factor interpretation the method's chronic, analyst-dependent bottleneck and sketched assigning factor meanings from high-loading variables' predicates—anticipating SFA's computable treatment of the items' language.

Coherence thus adds a response-free strand: agreement between item-meaning and response-covariance structure is intranetwork validity evidence (Cronbach & Meehl, 1955), the convergent–discriminant logic of the external aspect (Messick, 1995) frames divergence as information, and even a close match corroborates rather than constitutes validity

(Borsboom et al., 2004). One divergence source is well specified: Borsboom (2008) shows items can covary through direct causal relations rather than a common cause (panic's attack→worry→avoidance chain), so causal-network domains should show lower coherence, the causal arrow applying cleanly only to reflective models (Borsboom et al., 2004). Recovering structure from items also echoes Loevinger's structural validity—internal structure should parallel external manifestations (Clark & Watson, 2019).

Embeddings Predict Factor Structure and Item Correlations

A growing literature shows embedding similarity recovers inventories' a-priori factor structure without responses. Milano, Luongo, et al. (2025) embedded four scales (Big-Five markers, RIASEC, Humor Styles, DASS), and a principal component analysis of cosines loaded over 70% of items on their predicted construct, loadings correlating with response loadings; Casella et al. (2024) report the same for the Big Five. Guenole et al. (2025) formalize a *substitutability assumption*—an item's embedding stands in for its response vector—and run *pseudo-factor analysis* on the cosine matrix, recovering the five- and six-factor IPIP NEO-300 and HEXACO-240 structures.

The convergence comes with informative divergence, not equivalence: Guenole et al. (2025) find Tucker congruence often below conventional thresholds even as matched factors correlate and non-matched do not, while Milano, Luongo, et al. (2025) and Casella et al. (2024) report agreeableness and extraversion items bleeding together, mirroring their elevated behavioral correlation—semantic compression of distinct dimensions. Guenole et al. (2025) bound substitutability by training–response population similarity and intend pseudo-factor analysis as a pre-trial check—the response-free use case Milano, Luongo, et al. (2025) and Milano, Ponticorvo, and Marocco (2025) also state.

The most direct evidence predicts the empirical inter-item correlations. Hommel and Arslan (2025) fine-tuned a sentence transformer so cosine proximity estimates behavioral correlation, recovering out of sample, across 79 scales, interitem correlations at $r = .59$ and scale intercorrelations and reliabilities at $r = .84$. To overcome polarity—cosines rarely turn negative—they relabeled the SNLI corpus with signed magnitudes, training the model to return negative similarities for contradiction, letting a similarity matrix stand in for a covariance matrix (Bowman et al., 2015; Hommel & Arslan, 2025). This is not confined to fine-tuned encoders: Ravenda et al. (2025) found an item's strongest empirical partner among its three nearest semantic neighbors in 82% of Big-Five and 95% of DASS cases, an EFA reproducing the response structure. It even rivals expertise: Schoenegger et al. (2025) found AI outpredicted most individual humans, experts included, on item-pair correlations, though the residual gap cautions these approximate rather than replace empirical values. It is also partial: fitting confirmatory models to the cosine matrix against 31,697 VIA-IS-P respondents, Feraco and Toffalini (2025) found semantic and empirical correlations agreeing only moderately ($r = .67$)—every semantically fitting model fit the data, but only half of misfits truly misfit—so coherence is evidence, not identity, and SFA a complement to, not a substitute for, data collection.

Taxonomy, Jingle–Jangle, and Clinical Structure

Embedding analyses also address *taxonomic incommensurability*, the difficulty of comparing constructs across traditions that map theory onto measurement incompatibly (Wulff & Mata, 2025). Wulff and Mata (2025) placed 4,452 IPIP items, 459 scales, and 277 labels in a shared space, recovering known psychometric properties, and used the joint content–label embedding to automate *jingle–jangle* detection—a jingle when same-labeled scales differ semantically, a jangle when similar scales carry distinct labels—flagging 504 fallacies and a taxonomy with $\approx 75\%$ fewer labels. Wulff and Mata (2026) generalize this as scalable *conceptual engineering*, insisting it inform rather than replace expert judgment, since embeddings inherit training biases.

The logic extends to clinical items, themselves verbal symptom descriptions. Kambeitz et al. (2025) embedded four psychopathology scales (DASS, O-LIFE, AQ, Prodromal Questionnaire), found cosines tracking empirical item-pair correlations, and showed *k*-means clustering reproduced the established subdomains, the embedding and empirical partitions agreeing above chance by the adjusted Rand index (Hubert & Arabie, 1985)—better than chance, worse than the data. The contrast is sharpest against an external criterion: across 145 symptom items, Kojima et al. (2026) extracted factors from item-embedding cosines and from response polychorics, finding the general psychopathology factor nearly interchangeable (congruence = .97) yet specific factors diverging of non-semantic origin—insomnia and somatic items fused only in responses, the syndromic factor predicting diagnostic histories better than the semantic one. On the same corpus we revisit (Open-Source Psychometrics Project, 2018), Stanghellini et al. (2024) introduced *semantic loadings*—overlaps between cognitive-network communities of item concepts and exploratory-graph factors of responses—resolving emotional from physical anxiety facets EFA left merged.

The relation is construct-dependent: modeling 463 item pairs, Zhang and Qiu (2026) found cosine similarity predicted inter-item correlations but with large random-slope variance and a negative intercept–slope correlation—where baseline cohesion is high, similarity adds less. The same logic recovers social structure: Kmetty et al. (2021) factor-analyzed a similarity matrix over 234 occupation terms, extracting dimensions correlating $r \approx .71$–$.78$ with the survey-based socio-economic index, with informative divergences (health-care occupations under-ranked, an organizational-power dimension emerging from text alone). That word meaning is itself a measurable resource is long established: Buchanan et al. (2001) showed semantic-neighborhood measures from a co-occurrence space predict word recognition, with a precisely quantifiable advantage over feature-based measures.

Methods, Pipelines, and the Wider Applied Wave

A first cluster treats the encoding step. Because embeddings compress similarities toward the positive—so that pseudo-factor loadings do not differentiate reverse-keyed items

(Suárez-Álvarez et al., 2026)—SQuID subtracts the mean "questionnaire embedding" from each item, recovering a signed range without fine-tuning (Pellert et al., 2026). Fitting confirmatory models to mean-centered cosines, Pokropek (2026) shows via Monte Carlo that fit indices stay informative with six or more indicators but are met by chance with few. And Unique Variable Analysis detects local dependence via a regularized network and weighted topological overlap (Christensen et al., 2023), a screen that transfers to the semantic matrix because shared semantic reference drives such dependence.

A parallel line transports network-psychometric retention onto the embedding matrix. Exploratory Graph Analysis (EGA) estimates dimensionality by community detection (Garrido et al., 2025; Golino, 2026): benchmarking six facets, Garrido et al. (2025) found Kaiser's rule (Kaiser, 1960) and parallel analysis (Horn, 1965) overestimated dimensionality while EGA recovered it, near-perfectly after weighted-topological-overlap filtering (Christensen et al., 2023); Golino (2026) adds Dynamic EGA to optimize recovery. The geometry also supports response-free screening: Jung and Seo (2025) selected centroid-proximal IPIP markers by k-means, recovering a 30-item form matching response-based reductions (Open-Source Psychometrics Project, 2018); and Wang et al. (2026) clustered contextual embeddings by density, cutting items by $\approx 60\%$ across DASS, IPIP, and EPOCH while preserving structure—both relying on chance-corrected partition agreement (Hubert & Arabie, 1985) and projection (van der Maaten & Hinton, 2008). Hussain et al. (2024) run on-device models, showing item cosines track response correlations; and Low et al. (2024), flagging a "psychometric bias" judging models only by label prediction, name embedding-based factor analysis as open.

A generative wave drafts items rather than dissecting them: the AI-GENIE framework (Russell-Lasalandra et al., 2026) has a model write a pool pruned in silico (EGA, UVA, bootstrap EGA), reported to match representative samples for the best-performing models. Others dissect existing items: Huang et al. (2025) cluster item embeddings to flag redundancy, recovering in a preliminary study Grit–Conscientiousness overlap while keeping gratitude separate. A parallel strand uses models as respondents: Liu et al. (2025) calibrate LLM responses with item response theory, obtaining difficulties correlating with human values but narrow abilities—still a person-by-item matrix. Models also stand in at construction: Eberhardt et al. (2025) build an *LLM rating scale* from a local model's transcript ratings, reaching strong reliability; yet Grobelny et al. (2025) found a ChatGPT-4 adaptation matching the human one on structure but with lower internal consistency. Validation is then constitutive: Bunt et al. (2025) show a model's text coding earns trust only after iterative semantic, predictive, and content validity—an *abductive* use. Lim et al. (2025) screen items with simulated respondents on the Big Five, Schwartz values, and VIA; and Maharjan et al. (2025) scrutinized embeddings as trait measures across ∼1M Reddit posts. Urgency comes from a threat to response data: Westwood (2025) showed an autonomous agent passing 99.8% of attention checks, so text-only checks are immune to synthetic-respondent fraud.

A further line asks whether item language carries item-response-theory parameters. Synthesizing 37 articles, Peters et al. (2025) find text recovers *difficulty* usefully (r up to .87); Han et al. (2025) find *discrimination* largely text-resistant; and Ormerod (2026) recover it better by simulating graded-ability responses with fine-tuned generative models. The case is sharpened by critique: Uher (2025) argues statistics is not measurement and warns of a *cardinal error*—mistaking judgments of verbal statements for measurements of what they describe; yet this same critique supplies the divergence mechanism, items' inbuilt semantics accounting for much rated covariance. Finally, the tools carry their provenance: the sentence-similarity backbone derives from natural-language-inference training on the SNLI corpus (Bowman et al., 2015), which first enabled cosine-comparable embeddings (Reimers & Gurevych, 2019); visualization uses t-SNE (van der Maaten & Hinton, 2008); and interrogation uses semantic projection along antonymic anchors (Grand et al., 2022).

The Present Package

These threads converge on one opportunity. Item geometry encodes relational knowledge—ordering items between antonymic anchors recovers human feature judgements from the embedding alone, though intra-linguistically (Grand et al., 2022); embeddings thus capture the semantic geometry of a scale's language (Westbury, 2025), which for our purposes is its *wording* rather than the attribute itself. What has been missing is a single, general-purpose, response-free toolkit bringing the full discipline of exploratory factor analysis to that structure in one reproducible workflow.

`semanticfa` (Yanitski & Westbury, 2026) is that toolkit. Built on R (R Core Team, 2025) and interoperating with the factor-analytic and network-psychometric ecosystem (Golino & Christensen, 2026; Revelle, 2026), it implements SFA in five stages: *similarity construction* under keyed, keying-free, and natural-language-inference encodings; *retention* by multi-criterion consensus; *extraction* with Monte Carlo–calibrated diagnostics; *interpretation* by anchoring, semantic projection, congruence, and jingle–jangle screening; and response-free *refinement*. Semantic and behavioral solutions are compared by the chance-corrected adjusted Rand index (Hubert & Arabie, 1985).

Positioning clarifies what is new. The name *semantic factor analysis* was coined independently by Müller (2026) for a Word2Vec recovery of Big-Five structure from adjectives (Goldberg, 1990; Mikolov et al., 2013), each scaled by its mean response and partitioned. Our use denotes a distinct, broader method: a fully response-free factor analysis of the embedding similarity structure, not restricted to adjectives—answering, rather than committing, the cardinal error Uher (2025) identifies.

We demonstrate every exported function on the 50-item IPIP Big-Five Factor Markers, embedded with Qwen3-Embedding-8B (Qwen Team, 2025) and validated against a conventional factor analysis of 874,434 Open-Source Psychometrics respondents (Open-Source Psychometrics Project, 2018), asking whether consensus retention recovers the Big Five. We turn now to the method.

Method

Materials

The demonstration uses the 50-item International Personality Item Pool (IPIP) Big-Five Factor Markers, a public-domain inventory of ten short, first-person items (e.g., "I am the life of the party.") per domain: Extraversion, Emotional Stability, Agreeableness, Conscientiousness, and Openness/Intellect. The set operationalizes the five-factor structure Goldberg (1990) demonstrated across near-comprehensive pools of English trait adjectives—an ideal test bed for a semantic method, pairing one of psychometrics' most replicated target structures with brief, behaviorally concrete items whose meaning a language model can plausibly capture. The inventory ships in the package as `data(big5)`: item texts, codes (E1–O50), precomputed item embeddings (see the Embeddings subsection), the official ±1 scoring key—under which 18 of the 50 items are reverse-keyed, with Emotional Stability keyed in the Neuroticism direction—and the theoretical domain labels used in all analyses, tables, and figures below, which name the second domain Neuroticism (its keyed direction) and the fifth Openness (its short form). The scoring metadata enter only the explicitly keyed procedures (the `atomic_reversed` encoding and the scoring of the human responses); the primary similarity encoding is keying-free.

The human benchmark is the Open-Source Psychometrics Project's "Big Five Personality Test" raw-data release (Open-Source Psychometrics Project, 2018), an open administration of the same 50 items on a 5-point Likert scale in which 0 codes a skipped item. Of its 1,015,341 response records we removed every record with a missing response or one outside the 1–5 range, retaining $N = 874{,}434$ respondents; the same release has served as a validation corpus in earlier embedding work (Jung & Seo, 2025; Wulff & Mata, 2025). These responses serve *only* as the validation target—a conventional exploratory factor analysis (EFA) of their correlations against which the semantic solution is compared—and never enter the semantic analyses.

Embeddings

The main analyses use precomputed item embeddings from Qwen3-Embedding-8B (Qwen Team, 2025), a 4,096-dimensional text-embedding model. Each of the 50 item texts was embedded once, offline, with last-token pooling, no instruction prefix, and no normalization; the resulting 50×4096 matrix—with texts, codes, domain labels, and scoring key—ships as `data(big5)`, so the analyses need no Python or network access.

Label anchoring and semantic projection also require vectors for non-item words: the five construct names (*extraversion*, *neuroticism*, *agreeableness*, *conscientiousness*, *openness*) and the projection poles (*anxious*, *calm*, *solitary*, *sociable*). A script in the reproduction archive extracted these from a precomputed one-million-word lexicon embedded with the same model, placing them in the items' vector space as the cosine comparisons require.

The package's on-device default for new text is the smaller Qwen3-Embedding-0.6B (Qwen Team, 2025), yielding 50×1024 vectors for the same items, and backs the two demonstrations that must embed text live: the `sfa_embed()` re-embedding check, which compares fresh vectors with same-model offline reference vectors loaded from a NumPy `.npz` archive by `sfa_load_npz()` (the package's import route for external embeddings), and `sfa_item_fit()`'s vetting of candidate items that postdate the archives. Both run through `reticulate` and `sentence-transformers`, which downloads the model from the Hugging Face Hub on first use.

Finally, one similarity matrix is not embedding-based: `sfa_nli_matrix()` scores all item pairs with `cross-encoder/nli-deberta-v3-base`, a natural-language-inference (NLI) cross-encoder trained on SNLI-tradition corpora (Bowman et al., 2015); the next subsection describes its role.

The `semanticfa` Pipeline

The package (Yanitski & Westbury, 2026) exposes five stages as documented R functions (R Core Team, 2025).

Similarity. `sfa_similarity()` turns embeddings into an item-by-item similarity matrix under four encodings: `atomic` (cosines of L2-normalized vectors) and `atomic_reversed` (reverse-keyed embeddings times -1), the pseudo-factor encodings of Guenole et al. (2025), plus two keying-free encodings recovering the negative relations raw cosines suppress (notably for reverse-keyed items)—`squid`, which subtracts the questionnaire-mean embedding (Pellert et al., 2026), and `mean_centered_pearson`, which centers each vector across its dimensions into a genuine correlation matrix (Pokropek, 2026). A fifth, embedding-free route, `sfa_nli_matrix()`, scores pairs with the NLI cross-encoder above (Bowman et al., 2015) as entailment minus contradiction probability, the signed mapping of Hommel and Arslan (2025), and projects indefinite results (signed matrices need not be positive semi-definite), with notice, to a nearby positive-definite correlation matrix (eigenvalues floored, diagonal rescaled to one).

Retention. `sfa_parallel()` adapts parallel analysis (Horn, 1965) to embeddings—null eigenvalues come from similarity matrices of random unit vectors in the embedding dimension, no sample size involved—retaining by default factors above the 95th percentile of 100 null replicates. `sfa_nfactors()` reports the consensus of up to four criteria: parallel analysis; the latent-root-one rule (eigenvalues above one; Horn, 1965); minimum TEFI (Golino, 2026); and EGA (EBICglasso networks with Walktrap detection, via EGAnet; Golino & Christensen, 2026), which best recovers embedding dimensionality (Garrido et al., 2025). `sfa_dimselect()` adapts the embedding-depth optimization of Golino (2026), choosing the nested leading-coordinate depth maximizing $0.70 \times \text{NMI} - 0.30 \times \text{TEFI}_{\text{norm}}$.

Extraction and diagnostics. `sfa()` runs the pipeline end to end from raw text, embeddings, or a similarity matrix, with minimum-residual factoring and oblimin rotation (oblique, in the varimax tradition; Kaiser, 1958) via psych (Revelle, 2026). Fits report Kaiser–Meyer–Olkin (KMO) sampling adequacy, partition-based TEFI (Golino, 2026), root mean square residual (RMSR), common part accounted for (CAF), McDonald's omega per factor (empirical and best-matching theoretical sets), and, given domain labels, a factor-by-construct dominant average absolute loading (DAAL) matrix (Guenole et al., 2025). On the chance-baseline logic above, `calibrate = TRUE` re-derives RMSR, CAF, and TEFI reference distributions from null replicates with no shared semantic structure (Pokropek, 2026); `as_psych()` converts fits to `psych::fa` objects.

Interpretation. Following Wulff and Mata (2025), `sfa_anchor()` tabulates "belonging," a semantic loadings analogue, as cosine similarity to construct anchors (assigned-item centroids or label embeddings), flagging items weakest on their own construct or closer to a rival. `sfa_project()` applies the semantic projection of Grand et al. (2022) introduced above, scores rescaled so that 0 and 1 denote the two poles. `sfa_congruence()` compares a fit with a theoretical partition, human EFA, or correlation matrix on up to five metrics: factorwise Tucker's ϕ (`psych::factor.congruence`; Revelle, 2026); normalized mutual information (NMI) between partitions (Garrido et al., 2025; Golino, 2026); the adjusted Rand index (ARI) (Hubert & Arabie, 1985); a normalized Frobenius matrix similarity; and a disattenuated item-pair correlation (over split-half reliabilities, missing when either is negative). `sfa_jinglejangle()` screens scale sets for jingle–jangle fallacies—label versus item-content similarity per scale pair (Wulff & Mata, 2025, 2026)—flagging by default at $|\,\text{content} - \text{label}\,| \geq 0.20$.

Refinement. `sfa_redundancy()` flags locally dependent pairs by faithful UVA (Christensen et al., 2023)—absolute weighted topological overlap on an EBICglasso network of the similarity matrix (Golino & Christensen, 2026), recommended cutoff 0.25—suggesting removals that keep one item per cluster; a direct cosine criterion is a simpler alternative. `sfa_simplify()` builds response-free short forms from the items most representative of each construct's anchor or medoid (Jung & Seo, 2025; Wang et al., 2026), reporting retention and theory-recovery statistics against the original—candidates pending psychometric validation. `sfa_item_fit()` extends the anchoring logic of Wulff and Mata (2025) to vet newly written items against an existing fit: each is embedded with a named model and scored per construct on similarity to the construct name and its items, cross-loading gap to the second-best construct, and near-duplicates above a redundancy cutoff. `sfa_corplot()` draws the similarity matrix as a factor-grouped heatmap; `sfa_itemplot()` maps items to two dimensions by t-SNE (van der Maaten & Hinton, 2008), UMAP, principal components, or classical multidimensional scaling; and `plot()` methods add a scree plot with parallel-analysis overlay and a loadings heatmap.

Analysis Plan

One master script (`reproduce.R`, seed 42) ran the nine demonstration stages; its `R CMD BATCH` transcript supplies every number in the Results.

Similarity construction. The four embedding encodings came from the 50×4096 Qwen3-Embedding-8B matrix, the signed NLI matrix from the item texts; we inspected each encoding's off-diagonal range and, to probe reverse-keying, the similarities among Extraversion item E1, its reverse-keyed sibling E2, and same-keyed E5.

Live-embedding check. The cache was cleared, all 50 items re-embedded with the default Qwen3-Embedding-0.6B model via `sfa_embed()`, and per-item cosines to the `sfa_load_npz()` reference vectors summarized.

Retention. On the mean-centered Pearson matrix we ran parallel analysis (100 iterations, 95th percentile), the four-method `sfa_nfactors()` consensus (parallel, latent-root-one, TEFI, EGA), and the EGA-based depth optimization of `sfa_dimselect()`.

Main fit. `sfa()` fit the primary model on the keying-free `mean_centered_pearson` encoding, chosen before interpretation for its genuine correlation matrix and for matching theory and human data at least as well as every alternative in the encoding comparison. The factor number, first left to parallel analysis, was fixed at five throughout, with minimum-residual extraction, oblimin rotation, and the full diagnostic battery; `calibrate = TRUE` re-estimation (100 Monte Carlo replicates) supplied null reference distributions

(Pokropek, 2026), and `as_psych()` handed the fit to psych.

Interpretation. `sfa_anchor()` ran with both centroid and label anchors, the latter from the 8B construct-name vectors; `sfa_project()` placed all items on two bipolar 8B pole-word axes, stability (*anxious–calm*) and sociability (*solitary–sociable*) (Grand et al., 2022); theoretical-partition congruence was scored by NMI and ARI; and `sfa_jinglejangle()` screened the five subscales (8B item and label vectors, flag threshold 0.20).

Refinement. Redundancy screening used Unique Variable Analysis (threshold 0.25) and, as a check, the direct cosine criterion (threshold 0.85); `sfa_simplify()` built a short form (anchor selection, five items per construct, 50 to 25 items); and `sfa_item_fit()` vetted three new candidate items. Vetting embeds new text, so it ran in the default on-device model's 0.6B space, with the reference model deliberately fit *without* the scoring column (rationale in the Results); the keying-free similarity side is unaffected.

Model-size comparison. The mean-centered Pearson matrix and four-criterion retention comparison were recomputed from same-family 0.6B (1,024-dimensional) and 4B (2,560-dimensional) embeddings of the same items alongside the 8B used throughout.

Visualization. The script regenerates the mean-centered Pearson and NLI similarity heatmaps, item maps under all four projection methods, the scree plot with parallel-analysis overlay, and the loadings heatmap.

Human benchmark. After the Materials cleaning, the 18 reverse-keyed items were rescored ($6 - x$, the standard direction), yielding the keyed 50×50 Pearson matrix; the verbatim (unkeyed) matrix was kept for the keying-free comparisons. The human EFA used `psych::fa` (Revelle, 2026) on the keyed correlations (minimum residual, oblimin, $n = 874{,}434$), factors fixed a priori at the theoretical five (Goldberg, 1990); the Results add a conventional parallel-analysis recommendation for context.

Semantic–behavioral comparison. Coherence was operationalized at three levels. (a) `sfa_congruence()` gave structural congruence of the main fit with the human EFA on all five metrics, the disattenuated one computed separately at the item-pair level and reported as missing rather than suppressed where the keying flip leaves split-half reliability undefined (see the Results). (b) Item-pair agreement was the Pearson correlation of each of the five semantic matrices with the two human matrices over all 1,225 unique item pairs. (c) For the encoding comparison, a five-factor model was fit under every encoding (and the NLI matrix) and scored against theory, the human EFA (mean best-match $|\phi|$ congruence; oblimin factors are sign-indeterminate), the human matrices, and the internal diagnostics, on the metrics defined in the Results.

Open Practices

A self-contained reproduction archive, to be hosted on the Open Science Framework at `https://osf.io/XXXXX` [OSF link to be inserted], regenerates every number, table, and figure in the demonstration from raw inputs. It contains the master script (`reproduce.R`); a full-run console transcript (`reproduce.Rout`) for checking every command against its real output; the precomputed Qwen3-Embedding-8B item and auxiliary embedding archives with the auxiliary-vector generation script; results as an `.rds` object and CSV tables including the encoding comparison, congruence matrix, and both loading matrices; and the regenerated figures. The human response file is not redistributed where licensing forbids it; the script documents the public-release download (Open-Source Psychometrics Project, 2018) and accepts its location via an environment variable. Only two steps require Python or network access—the NLI matrix and the live-embedding demonstrations, which fetch models from the Hugging Face Hub on first run; all factor analyses, retention methods, diagnostics, congruence metrics, refinement analyses, and figures are pure R and run offline.

The reported run used R 4.5.0 (R Core Team, 2025) with semanticfa 0.1.0 (Yanitski & Westbury, 2026), psych 2.6.5 (Revelle, 2026), EGAnet 2.4.1 (Golino & Christensen, 2026), GPArotation 2026.4-1, Rtsne 0.17, uwot 0.2.4, and reticulate 1.43.0, with seed 42. The package is on CRAN, with development sources and issue tracker at `https://github.com/devon7y/semanticfa`; a vignette walks every exported function through a second, independent worked example.

Results

Embeddings and Similarity Encodings

Verbatim transcript excerpts (abridged where noted) trace the analyst workflow, starting with the bundled inventory and its Qwen3-Embedding-8B vectors:

```
> data(big5)   # 50 IPIP Big-Five Factor Markers + bundled Qwen3-Embedding-8B
> str(big5, max.level = 1)
List of 5
 $ items     : chr [1:50] "I am the life of the party." "I don't talk a lot."
   "I feel comfortable around people." "I keep in the background." ...
 $ codes     : chr [1:50] "E1" "E2" "E3" "E4" ...
 $ factors   : chr [1:50] "Extraversion" "Extraversion" "Extraversion"
   "Extraversion" ...
 $ scoring   : num [1:50] 1 -1 1 -1 1 -1 1 -1 1 -1 ...
 $ embeddings: num [1:50, 1:4096] 0.0361 0.0193 0.0184 0.0269 0.0081 0.0227
   0.0204 0.0234 0.0214 0.0189 ...
  ..- attr(*, "dimnames")=List of 2
> E8 <- big5$embeddings    # 50 x 4096
```

`sfa_similarity()` derives all four Methods-defined encodings from one matrix; `atomic_reversed` sign-flips the 18 reverse-keyed embeddings, and `squid` follows Pellert et al. (2026):

```
> sim_at  <- sfa_similarity(E8, "atomic",
+                            factors = big5$factors, codes = big5$codes)
> sim_ar  <- sfa_similarity(E8, "atomic_reversed", scoring = big5$scoring,
+                            factors = big5$factors, codes = big5$codes)
> sim_sq  <- sfa_similarity(E8, "squid",
+                            factors = big5$factors, codes = big5$codes)
> sim_mcp <- sfa_similarity(E8, "mean_centered_pearson",
+                            factors = big5$factors, codes = big5$codes)
> enc_range
                        min   max
atomic                0.394 0.938
atomic_reversed      -0.777 0.938
squid                -0.267 0.827
mean_centered_pearson  0.394 0.938
```

Off-diagonals confirm raw cosines' narrow positive band (Hommel & Arslan, 2025; Pellert et al., 2026): .394–.938 over all 1,225 pairs, even opposite-keyed ones (Figure 1, left); negatives must be manufactured (sign-flipping, −.777; `squid`, −.267), whereas `mean_centered_pearson` matched the raw-cosine range to three decimals.

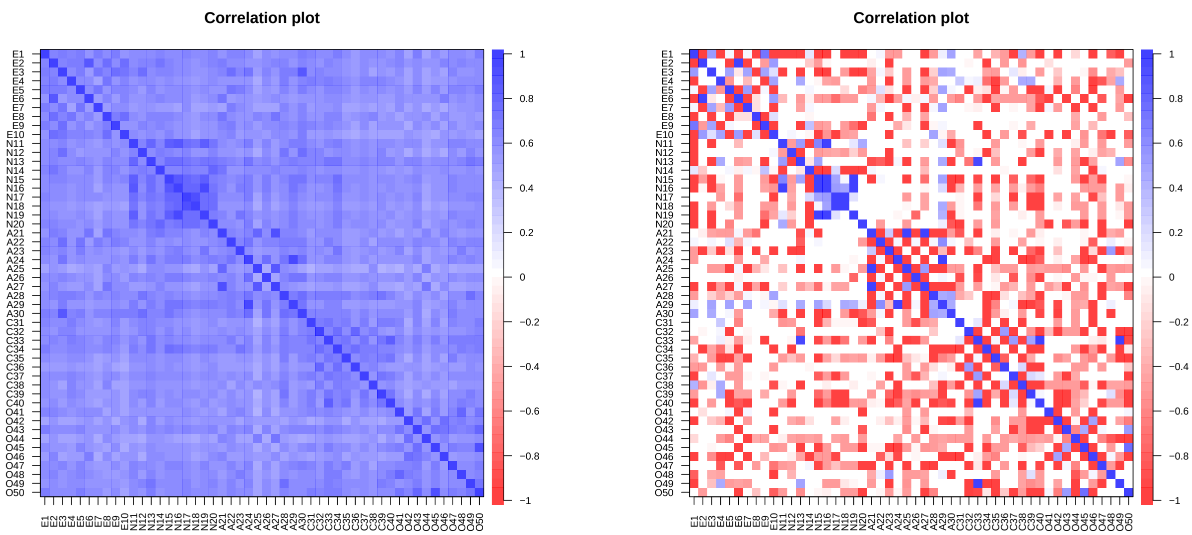

Figure 1

Left: cosine similarity (`atomic` encoding) among the 50 IPIP Big-Five Factor Marker items, grouped by domain (`sfa_corplot()`). All pairwise similarities are positive (range = .394–.938), with darker within-domain blocks (most visibly the Neuroticism items N15–N19) superimposed on a uniformly positive field. Right: signed NLI similarity (entailment minus contradiction) among the 50 items. In contrast to the left panel, contradictions between forward- and reverse-keyed items within a domain produce vivid negative (red) bands and within-block checkerboard structure.

For a signed alternative, `sfa_nli_matrix()` scores ordered pairs with `cross-encoder/nli-deberta-v3-base` in the SNLI tradition (Bowman et al., 2015), symmetrizing entailment minus contradiction (cf. Hommel & Arslan, 2025):

```
> t_nli <- system.time(
+   sim_nli <- sfa_nli_matrix(big5$items)   # cross-encoder/nli-deberta-v3-base
```

```
+ )
NLI matrix: 50 x 50 in 78 s
> round(sim_nli[c("E1", "E2", "E5"), c("E1", "E2", "E5")], 2)
      E1    E2    E5
E1  1.00 -1.00  0.03
E2 -1.00  1.00 -0.98
E5  0.03 -0.98  1.00
> round(sim_ar[c("E1", "E2", "E5"), c("E1", "E2", "E5")], 2)
      E1    E2    E5
E1  1.00 -0.63  0.67
E2 -0.63  1.00 -0.65
E5  0.67 -0.65  1.00
```

NLI scores E1 and its reverse-keyed sibling E2 at -1.00, near-perfect contradiction, versus $-.63$ under sign-flipping, which reverses vectors without making opposite paraphrases maximally dissimilar (Figure 1, right).

The live-embedding check (Methods) cleared the cache and re-embedded the 50 items with the default Qwen3-Embedding-0.6B:

```
> REF06_NPZ <- "data/Big5_items_0.6B.npz"
> ref06 <- sfa_load_npz(REF06_NPZ)
> ref06
Loaded embeddings (sfa_embeddings)
  Items: 50  |  Dimensions: 1024
  Metadata: codes, items, factors
  Factors: Extraversion, Neuroticism, Agreeableness, Conscientiousness, Openness
  Codes:   E1 E2 E3 E4 E5 E6 E7 E8 ...
> sfa_clear_cache()
> t_emb <- system.time(emb06 <- sfa_embed(big5$items))
sfa_embed: 50 x 1024 in 189.3 s
> print(summary(round(cos_live_ref, 4)))
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.9999  0.9999  0.9999  0.9999  0.9999  0.9999
```

Every live re-embedding matched its same-model offline reference at cosine $\geq .9999$, reproducing the archived output on-device.

Factor Retention

With no respondents to permute, `sfa_parallel()` adapts parallel analysis (Horn, 1965), retaining factors above the 95th percentile of random-embedding null eigenvalues:

```
> pa <- sfa_parallel(sim_mcp, E8)
> pa
Embedding-adapted parallel analysis
  Suggested factors: 5
  Embedding dimension: 4096
  Iterations: 100
  Percentile: 95
```

`sfa_nfactors()` adds three criteria: eigenvalues greater than one (Kaiser, 1960), TEFI (following Golino, 2026), and EGA, validated for embeddings by Garrido et al. (2025):

```
> nf <- sfa_nfactors(sim_mcp, embeddings = E8,
+                    methods = c("parallel", "kaiser", "TEFI", "EGA"))
> nf
Factor retention analysis (embedding-adapted)

  Method      n_factors
  parallel    5
  kaiser      6
  TEFI        2
  EGA         5
  ------------------------
  Consensus   5

Eigenvalues: 29.63   1.92   1.74   1.48   1.27   1.01   0.93   0.88   0.78   0.70  ...
```

Parallel analysis and EGA indicated five factors, Kaiser six, TEFI two; the consensus of five matches the theoretical Big Five, and the dominant first eigenvalue (29.63), the general positivity of embedding similarities, precedes a shallow elbow (Figure 2).

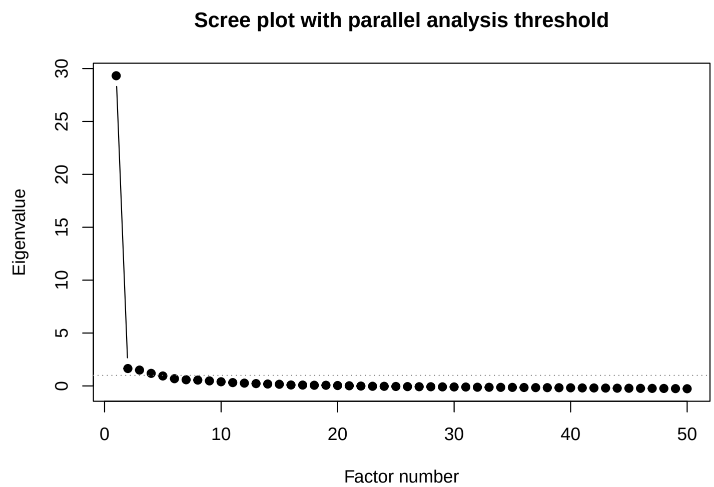

Figure 2

Scree plot of the mean-centered Pearson similarity matrix with the embedding-adapted parallel-analysis threshold (dotted line), produced by `plot(fit, type = "scree")`*. The first eigenvalue (29.63) dwarfs the rest; five eigenvalues clear the parallel-analysis threshold.*

A second question is embedding-specific: how many of the 4,096 coordinates carry structural signal? `sfa_dimselect()` adapts the embedding-depth optimization of Golino (2026), scoring swept truncations by structural recovery (NMI) weighted against entropy fit (TEFI):

```
> ds <- sfa_dimselect(E8, factors = big5$factors,
+                     encoding = "mean_centered_pearson")
> ds
EGA-based embedding-dimension selection
  Method: EGA depth optimization (adapts Golino 2026)
  Full embedding dim: 4096
  Depths evaluated:   148 (range 3-4096)
  Criterion:          0.70*NMI - 0.30*TEFI (normalized)
  Optimal depth:      1431 of 4096 coordinates
    at optimum: n_dim=5  NMI=0.704  TEFI=-23.291
> cat("sfa_dimselect elapsed:", round(t_ds["elapsed"], 1), "s\n")
sfa_dimselect elapsed: 13.6 s
```

The optimum—roughly the first third, 1,431 of 4,096 coordinates—recovered five

EGA dimensions (NMI = .704), the five-factor answer by a different route; the 148-depth sweep took 13.6 s.

Model Size and Structural Clarity

A natural question is how much the embedding model's size matters. To show the gradient directly, the same 50 items were embedded with three sizes of the same model family—Qwen3-Embedding-0.6B (1,024 dimensions), 4B (2,560), and 8B (4,096)—and the mean-centered Pearson similarity matrix was built from each. Figure 3 shows the three matrices side by side, items grouped by domain: the five diagonal blocks sharpen visibly with model size, with the 0.6B matrix showing the haziest block boundaries and the 8B matrix the crispest separation of within-domain from between-domain similarity. Retention sharpens correspondingly. Under the 0.6B and 4B embeddings the criteria disagree—parallel analysis retains four factors, EGA six, for a modal consensus of six—whereas under the 8B embeddings parallel analysis and EGA converge on the theoretical five, and the consensus is five (the eigenvalue-greater-than-one rule reports six and TEFI two at every size). Larger models of the same family thus yield semantic structure that is both visibly cleaner and dimensionally closer to theory, which is why the demonstration's main analyses use the 8B embeddings while the package's on-device default remains the lighter 0.6B model.

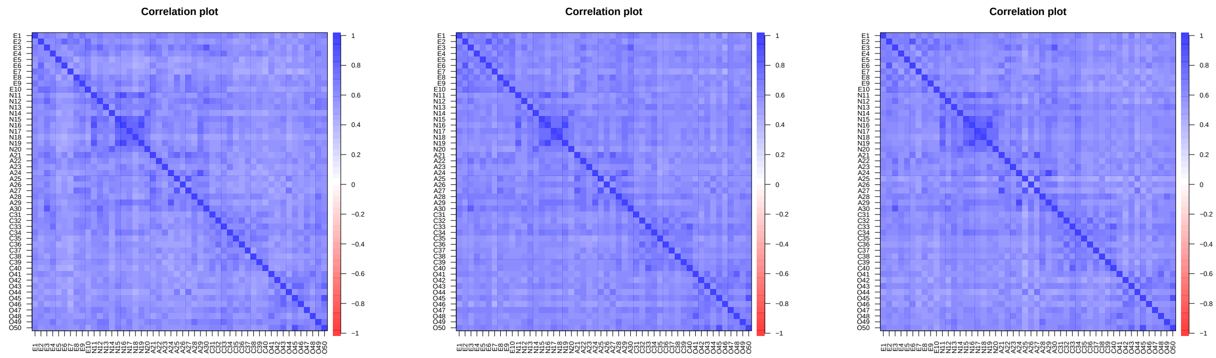

Figure 3

Mean-centered Pearson similarity matrices of the 50 IPIP items, grouped by domain, for Qwen3-Embedding-0.6B (left), 4B (center), and 8B (right). The domain block structure grows progressively crisper with model size.

Model Fit and Diagnostics

`sfa()` runs the whole pipeline in one call; automatic retention chose five factors, fixed as `nfactors = 5`, and the warning confirms the keying-free encoding ignores the scoring column.

```
> fit_auto <- sfa(big5_df, embeddings = E8, encoding = "mean_centered_pearson")
Warning message:
'scoring' has reverse-keyed items but is ignored for encoding =
"mean_centered_pearson": this method is keying-free by design. Use
"atomic_reversed" for keyed sign-flipping.
> cat("Auto-retained factors:", fit_auto$factors, "\n")
Auto-retained factors: 5
> fit <- sfa(big5_df, embeddings = E8, encoding = "mean_centered_pearson",
+            nfactors = 5)
> fit
Semantic Factor Analysis
  Encoding: mean_centered_pearson
  Embedding dim: 4096
  Factors:5 (minres + oblimin)

Diagnostics:
  KMO:  0.971 (marvelous - higher is better)
```

```
  TEFI: -27.7059 (lower is better)
  RMSR: 0.0323 (good - lower is better)
  CAF:  0.4120 (marginal - higher is better)

[item-by-factor loadings omitted; reported in full in the supplement]

                  MR3   MR1   MR2   MR5   MR4
SS loadings     6.421 5.330 4.111 3.193 2.736
Proportion Var 0.128 0.107 0.082 0.064 0.055
Cumulative Var 0.128 0.235 0.317 0.381 0.436

Factor correlations (Phi):
      MR3   MR1   MR2   MR5   MR4
MR3 1.000 0.685 0.606 0.577 0.539
MR1 0.685 1.000 0.538 0.582 0.555
MR2 0.606 0.538 1.000 0.465 0.446
MR5 0.577 0.582 0.465 1.000 0.398
MR4 0.539 0.555 0.446 0.398 1.000
```

KMO was .971, RMSR .032, TEFI −27.71, and CAF .412; substantial factor correlations ($\phi = .40$–$.69$) again reflect the similarities' shared positive component. `summary(fit)` adds the per-factor McDonald's omegas:

```
McDonald's omega per factor:
 factor omega_assigned omega_theoretical matched_theoretical
    MR3          0.908             0.902         Neuroticism
    MR1          0.863             0.850   Conscientiousness
    MR2          0.830             0.560       Agreeableness
    MR5          0.709             0.583        Extraversion
    MR4          0.704             0.732            Openness
```

The solution recovers the Big Five: Neuroticism anchors MR3, Conscientiousness MR1, Openness MR4; MR2 and MR5 split Agreeableness and Extraversion items into warmth/other-orientation and gregariousness factors (Figure 4; Table S1). Communalities ranged from .566 (A23) to .905 (N16); the assigned–theoretical omega gaps—MR2 .830 vs. .560, MR5 .709 vs. .583—quantify this mixing: internally coherent factors that cross the theoretical boundary.

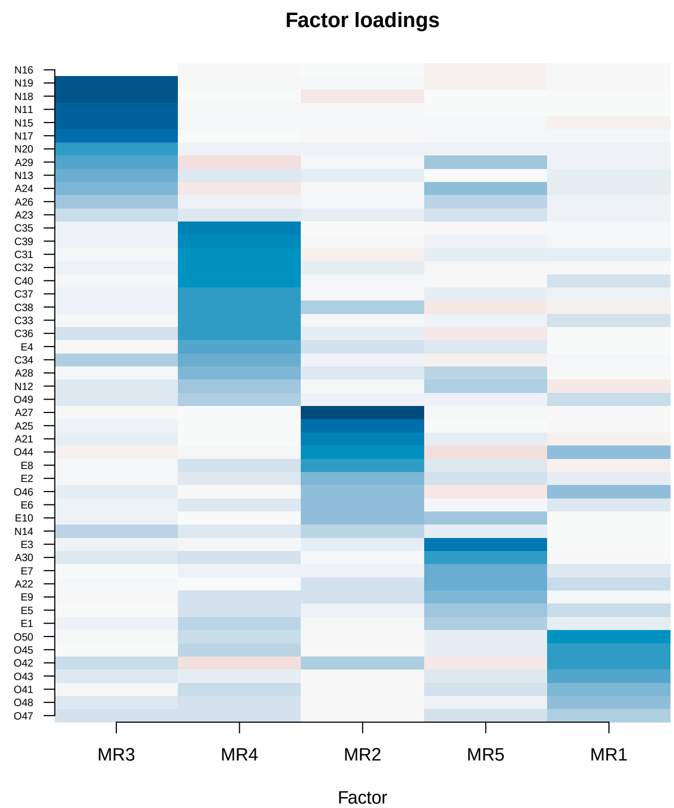

Figure 4

Item-by-factor loading heatmap for the five-factor semantic solution, produced by `plot(fit, type = "loadings")`. *Items are sorted by their dominant factor; MR3 = Neuroticism, MR1 = Conscientiousness, MR2 and MR5 = Agreeableness/Extraversion blends, MR4 = Openness.*

Conventional thresholds need not transfer to embedding-derived matrices (Pokropek, 2026), so `calibrate = TRUE` draws Monte Carlo reference distributions for RMSR, CAF, and TEFI from data-matched null embeddings (100 replicates, 54.7 s):

```
> fit_cal <- sfa(big5_df, embeddings = E8, encoding = "mean_centered_pearson",
+                 nfactors = 5, calibrate = TRUE, calibrate_iter = 100)
> cat("calibrate=TRUE elapsed:", round(t_cal["elapsed"], 1), "s\n")
calibrate=TRUE elapsed: 54.7 s
> fit_cal$calibration
[lists of 100 null RMSR, CAF, and TEFI values; omitted]
```

Across the 100 nulls, RMSR ranged from .0119 to .0134, CAF from .501 to .532, and TEFI from −21.66 to −17.82; the observed TEFI (−27.71) undercut every null—a more entropically organized five-factor partition—while the observed RMSR (.032) and CAF (.412) fell above and below their ranges: the real similarities carry residual structure beyond five factors that nulls lack, an index-by-index recalibration that raw thresholds would obscure. `as_psych(fit)` lets `psych::fa.sort()` and `psych::factor.congruence()` treat the fit like a response-based solution.

Interpretation Tools

After Wulff and Mata (2025), `sfa_anchor()` scores each item's cosine to every construct centroid or construct-name embedding, flagging items weak or beaten on their own construct:

```
> anc <- sfa_anchor(fit, anchor = "centroid")
> anc
Semantic anchoring
  Method: Wulff & Mata (2025)
  Anchor: centroid
  Constructs: 5 | Items: 50

[50 x 5 item-by-construct cosine matrix omitted]

  Weakest own-construct similarity (review/removal candidates):
    N12          Neuroticism        own=0.69   <-- higher on Extraversion (0.72)
        "I am relaxed most of the time."
    O44          Openness           own=0.69
        "I am not interested in abstract ideas."
    A23          Agreeableness      own=0.71   <-- higher on Extraversion (0.71)
        "I insult people."
    A25          Agreeableness      own=0.71
        "I am not interested in other people's problems."
    [list abridged]
```

Centroid anchoring mostly flags reverse-keyed items worded nearer another domain—N12 sits nearer the Extraversion centroid (.72) than its own (.69); label anchoring (same Qwen3-Embedding-8B space) flagged an overlapping set, A25 lowest (own .36 vs. .37 on Neuroticism), then E6, A27, C36, N12, and C32.

`sfa_project()` applies the semantic projection of Grand et al. (2022), placing items on bipolar axes via antonym-pole difference vectors from the same 8B space:

```
> prj <- sfa_project(fit,
+                    axes = list(stability   = c(low = "anxious",  high = "calm"),
+                                sociability = c(low = "solitary", high = "sociable")),
+                    pole_embeddings = poles)
> prj
Semantic projection onto 2 axis/axes
  Method: Grand et al. (2022)
  Scale: 0 = low pole, 1 = high pole

Axis 'stability'  [anxious <-> calm]
  lowest: N13=0.28  N11=0.33  N16=0.37  N19=0.38  N15=0.42
  highest: N12=0.82  A30=0.72  E3=0.65  E4=0.64  C37=0.62

Axis 'sociability'  [solitary <-> sociable]
  lowest: O46=0.37  A27=0.38  A21=0.38  N20=0.39  E4=0.41
  highest: E3=0.82  E7=0.81  A22=0.68  E5=0.68  A30=0.65
```

The projections track content, not scoring key: Neuroticism items fill the anxious pole, yet the most calm-pole item (.82) is again reverse-keyed N12, while party- and conversation-themed E3 and E7, and other-oriented A22 and A30, top the sociable pole.

Versus the theoretical Big Five assignment, `sfa_congruence()` scored agreement as moderate by NMI and weak by the chance-corrected ARI (Hubert & Arabie, 1985):

```
> cong_theory <- sfa_congruence(fit, target = big5$factors,
+                               metrics = c("nmi", "ari"))
> cong_theory
Factor structure congruence

  NMI:            0.527 (moderate - higher is better)
  ARI:            0.402 (weak - higher is better)
```

`sfa_jinglejangle()` compares label–label with content–content similarities (Wulff

& Mata, 2025, 2026), here treating the five 10-item subscales as separate scales (8B item and label vectors):

```
> jj <- sfa_jinglejangle(sub_items, item_embeddings = sub_emb,
+                         label_embeddings = jj_lab_emb)
> jj
Jingle-jangle detection across 5 scales
  Method: Wulff & Mata (2025, 2026)
  Flag threshold |content - label| >= 0.20

  No jingle/jangle pairs flagged.
```

Within this single, well-constructed instrument, label and content similarities align; its habitat is cross-instrument screening, where such fallacies are common (Wulff & Mata, 2025).

Response-Free Refinement

Three functions refine a scale before any respondent is recruited. `sfa_redundancy()` detects locally dependent item pairs with Unique Variable Analysis—wTO on an EBICglasso network (Christensen et al., 2023):

```
> red_uva <- sfa_redundancy(fit)                    # UVA / wTO (EGAnet)
> red_uva
Redundant-item detection
  Method: Christensen et al. (2023)
  Measure: wto | threshold: 0.25
  Redundant pairs: 11 | redundant clusters: 10

  Top redundant pairs:
    A24         ~ A29         overlap=0.462
    N17         ~ N18         overlap=0.402
    O45         ~ O50         overlap=0.374
    N16         ~ N19         overlap=0.361
    E2          ~ E6          overlap=0.339
    A25         ~ A27         overlap=0.331
    A21         ~ A27         overlap=0.320
    C32         ~ C36         overlap=0.309
    C33         ~ C40         overlap=0.286
```

```
  O41        ~ O48        overlap=0.268

 Suggested removals (keep one per cluster): E6, N19, N18, A25, A27, A29,
   C36, C40, O41, O44, O45
```

Every displayed pair among the eleven (in ten clusters) joins items from one domain; the simpler cosine criterion (threshold 0.85) flagged a largely overlapping set of seven pairs in five clusters, led by N16 ∼ N19 (.938) and A24 ∼ A29 (.933)—near-paraphrases, the "cheating by repeating" that inflates internal consistency without broadening coverage (Christensen et al., 2023).

`sfa_simplify()` builds candidate short forms from the items nearest each construct anchor (Jung & Seo, 2025; Wang et al., 2026):

```
> short <- sfa_simplify(fit, target_n = 5, method = "anchor")   # 50 -> 25 items
> short
Scale simplification (response-free)
  Method: centroid/medoid item selection (in the spirit of Wang et al. 2026;
    Jung & Seo 2025)
  Note: candidate short form -- validate psychometrically before use
  Selection: anchor | groups: theoretical | target_n = 5 per group
  Items: 50 -> 25
  Factors retained (parallel analysis): 5 -> 3
  Structure recovery vs theory (NMI / ARI):
    full:    NMI=0.527  ARI=0.402
    reduced: NMI=0.810  ARI=0.571
  Items kept per construct:
    Extraversion        5
    Neuroticism         5
    Agreeableness       5
    Conscientiousness   5
    Openness            5

  Dropped 25 item(s); see $drop. Reduced fit in $reduced_fit.
> short$keep
 [1] "E2"  "E3"  "E5"  "E8"  "E10" "N11" "N15" "N16" "N17" "N19" "A21" "A22"
[13] "A24" "A28" "A29" "C33" "C34" "C35" "C37" "C40" "O42" "O43" "O45" "O48"
[25] "O50"
```

Halving the scale to 25 items *improved* theoretical-partition recovery (NMI .527 to .810; ARI .402 to .571): pruning semantically peripheral items sharpens boundaries the full pool blurs. Parallel analysis now suggested three factors; per the printed caveat, a semantic short form is a candidate for psychometric validation, not a finished instrument.

`sfa_item_fit()` vets newly written items against an existing fit, extending Wulff and Mata (2025). Live embedding confines the demonstration to the default-model (0.6B) space; the reference fit omits the scoring column for the *anti-topic* reasons given in the Method section.

```
> fit06 <- sfa(big5_df[, c("code", "item", "factor")],
+              embeddings = ref06$embeddings,
+              encoding = "mean_centered_pearson", nfactors = 5)
> cand <- c("I make friends easily.",
+           "I am the life of every party.",
+           "I rarely feel anxious or depressed.")
> ifit <- sfa_item_fit(fit06, cand, model = "Qwen/Qwen3-Embedding-0.6B",
+                      redundancy_cutoff = 0.85)
> ifit
Candidate item fit
  Method: Wulff & Mata (2025), extended

Candidate 1: "I make friends easily."
  [similarity-by-construct table omitted]
  Best construct : Extraversion
  Cross-loading  : gap to 2nd (Neuroticism) = 0.01  ->  ambiguous -
    cross-loads with Neuroticism
  Strength       : about typical for Extraversion (items 0.74 vs avg 0.75)
  Redundancy     : closest item E3 "I feel comfortable around people."
    (0.78) -> distinct
  Verdict        : cross-loads (Extraversion vs Neuroticism) - poor
    discrimination

Candidate 2: "I am the life of every party."
  [similarity-by-construct table omitted]
  Best construct : Extraversion
  Redundancy     : closest item E1 "I am the life of the party." (0.90)
    -> near-duplicate
  Verdict        : redundant (near-duplicate of an existing item)
```

```
Candidate 3: "I rarely feel anxious or depressed."
  [similarity-by-construct table omitted]
  Best construct : Neuroticism
  Strength       : weaker than average for Neuroticism (items 0.75 vs
    avg 0.81)
  Verdict        : weak match - fits Neuroticism worse than its existing
    items (may not belong to this scale)
```

All three planted candidates drew the intended verdicts: cross-loading (Extraversion vs. Neuroticism gap .01), near-duplication of E1 (cosine .90, above the .85 cutoff), and a weaker match to its best construct than that construct's existing items.

Visualization

Beyond the `sfa_corplot()` similarity heatmaps (Figure 1, left and right panels) and the `plot()` scree and loading plots (Figures 2 and 4), `sfa_itemplot()` maps items into two dimensions by t-SNE (van der Maaten & Hinton, 2008), UMAP, PCA, or classical MDS:

```
> for (m in c("tsne", "umap", "pca", "mds")) {
+   sfa_itemplot(fit, method = m, seed = 42, legend = (m == "tsne"))
+ }
```

In all four projections (Figure 5), Neuroticism, Conscientiousness, and Openness items form compact, separated neighborhoods, whereas Extraversion and Agreeableness items interleave in a shared sociability region—a geometric preview of the factor solution's MR2/MR5 blending and the congruence results that follow.

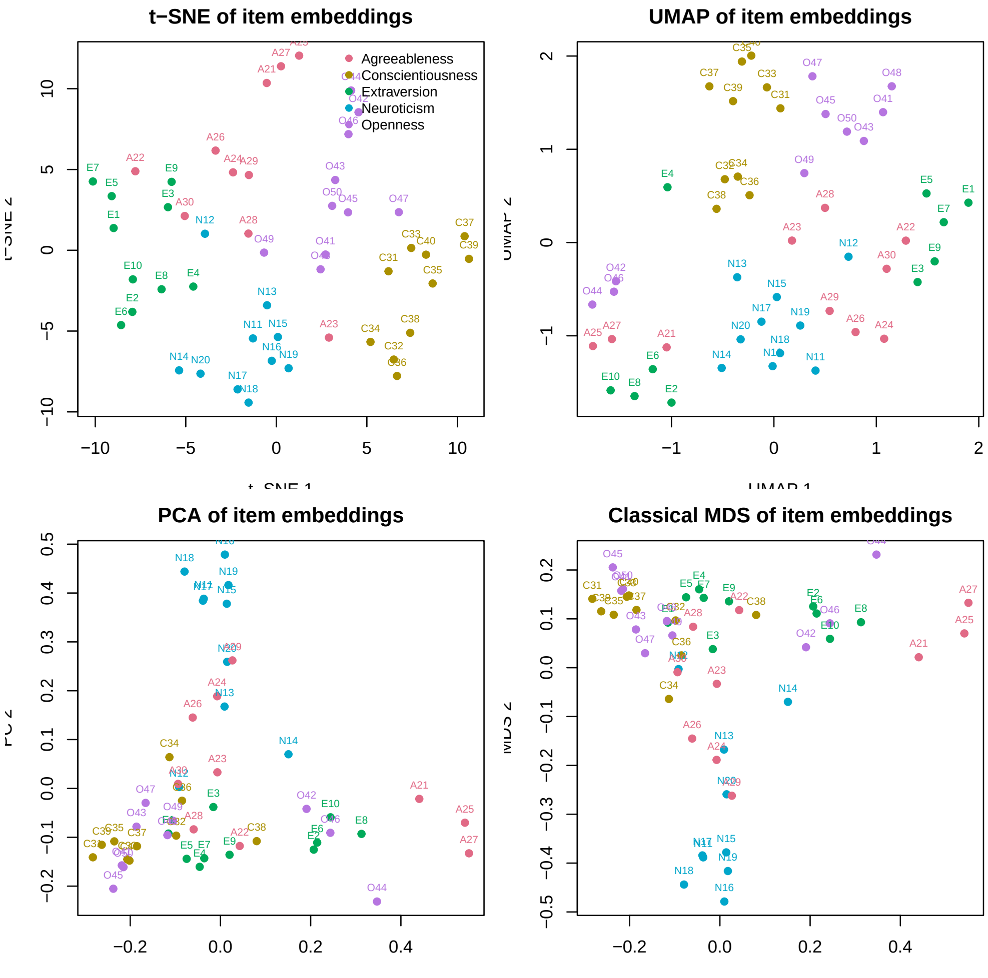

Figure 5

Two-dimensional item maps of the 50 item embeddings produced by `sfa_itemplot()`*: t-SNE (top left), UMAP (top right), PCA (bottom left), and classical MDS (bottom right), with items colored by theoretical domain. Neuroticism, Conscientiousness, and Openness items cluster tightly in all four projections; Extraversion and Agreeableness items interleave.*

Semantic–Behavioral Coherence

Cleaning the human benchmark (see Method) left 874,434 of 1,015,341 respondents, the 18 reverse-keyed items rescored ($6 - x$) for keyed analyses:

```
> keep <- rowSums(human_raw < 1 | human_raw > 5 | is.na(human_raw)) == 0
> human <- human_raw[keep, ]
> cat("Respondents after cleaning:", n_human, "of", nrow(human_raw), "\n")
```

```
Respondents after cleaning: 874434 of 1015341
> human_fa <- psych::fa(R_human, nfactors = 5, n.obs = n_human,
+                        rotate = "oblimin", fm = "minres")
```

Five factors were fixed a priori (Goldberg, 1990); parallel analysis suggested up to ten—at $N = 874{,}434$ even trivial factors clear noise—but five reproduced the Big Five with simple structure (Table S2).

Factor-Level Congruence

`sfa_congruence()` takes a fitted `psych` model as the target:

```
> cong_human <- sfa_congruence(fit, target = human_fa,
+                               metrics = c("tucker", "nmi", "ari", "frobenius"))
> cong_human
Factor structure congruence

Tucker phi (factor-by-factor):
     MR4   MR1  MR3  MR2  MR5
MR3 0.03  0.80 0.30 0.13 0.12
MR1 0.27  0.11 0.10 0.84 0.28
MR2 0.41 -0.01 0.52 0.05 0.27
MR5 0.64  0.07 0.52 0.11 0.16
MR4 0.18  0.02 0.17 0.14 0.92

  NMI:            0.527 (moderate - higher is better)
  ARI:            0.402 (weak - higher is better)
  Frobenius:      0.664 (moderate - higher is better)
```

Each semantic factor found a distinct human counterpart (Table 1): Openness, Conscientiousness, and Neuroticism strongly ($\phi = .92$, .84, .80); Extraversion (.64) and Agreeableness (.52) weakly, the former matching human Agreeableness nearly as well (.52). Overall, $\bar{\phi} = .744$, NMI = .527, ARI = .402, and normalized Frobenius similarity .664 between the correlation structures.

Table 1

Tucker congruence coefficients between the semantic factors (rows) and the human-response factors (columns)

	Human factor				
Semantic factor	E (MR4)	N (MR1)	A (MR3)	C (MR2)	O (MR5)
N (MR3)	.03	**.80**	.30	.13	.12
C (MR1)	.27	.11	.10	**.84**	.28
A (MR2)	.41	−.01	**.52**	.05	.27
E (MR5)	**.64**	.07	.52	.11	.16
O (MR4)	.18	.02	.17	.14	**.92**

Note. *E = Extraversion; N = Neuroticism; A = Agreeableness; C = Conscientiousness; O = Openness. MR labels are the rotation-order factor names from each solution. Bold values mark each semantic factor's best-matching human factor (mean matched* $\bar{\phi} = .744$*). Semantic solution: minres/oblimin on the mean-centered Pearson similarity matrix; human solution: minres/oblimin on keyed response correlations,* $N = 874{,}434$.

Comparing the Encodings

Table 2 extends this to every encoding's five-factor solution; the NLI matrix, supplied via `similarity =`, was indefinite (minimum eigenvalue −14.1), requiring positive-definite projection (see Method).

Table 2

Encoding comparison: congruence with theory, with the human factor solution, and with human item-pair correlations, plus fit diagnostics

	Theory		Human data				Diagnostics			
Encoding	NMI	ARI	$\bar{\phi}$	r_{keyed}	r_{raw}	r_{dis}	KMO	TEFI	RMSR	CAF
atomic	.527	.402	.744	.434	.456	.707	.971	−27.72	.032	.412
atomic_reversed	.527	.402	.438	.137	.444	—	.971	−27.72	.032	.412
squid	.391	.272	.636	.567	.490	1.000	.022	−44.39	.072	.996
mean_centered_pearson	.527	.402	.744	.434	.456	.707	.971	−27.71	.032	.412
nli	.548	.448	.310	.063	.561	—	.551	−54.15	.094	.688

Note. *NMI = normalized mutual information and ARI = adjusted Rand index of the five-factor item clustering against the theoretical Big Five partition; $\bar{\phi}$ = mean Tucker congruence after matching each semantic factor to its best human factor; r_{keyed} / r_{raw} = correlation between the encoding's 1,225 item-pair similarities and the human inter-item correlations computed from keyed and raw responses, respectively; r_{dis} = disattenuated congruence against the keyed human correlations. Dashes mark cases where the disattenuated metric is undefined and reported as NA because the similarity matrix's split-half reliability is not positive: the keying flip of* `atomic_reversed` *imposes a checkerboard sign pattern (the signed NLI matrix behaves analogously). KMO, TEFI, RMSR, and CAF are the fit diagnostics of each encoding's five-factor solution.*

Three patterns stand out. First, `atomic_reversed`—an anti-topic vector, not reverse-scoring's analogue (see Discussion)—aligned worst among cosines with keyed correlations (r_{keyed} = .137 vs. .434 unflipped) and ordinarily only with raw (r_{raw} = .444).

Second, NLI mirrors the cosines: best on the theoretical partition (NMI = .548, ARI = .448) and on raw pairwise correlations (r_{raw} = .561), worst on loading shape ($\bar{\phi}$ = .310), cosine the reverse ($\bar{\phi}$ = .744, r_{raw} = .456)—directional entailment versus graded geometry (see Discussion).

Third, SQuID won keyed pair-level agreement ($r_{\text{keyed}} = .567$; r_{dis} saturated at 1.000) yet was essentially unfactorable (KMO = .022), its differencing keeping pairwise relations but stripping extractable common variance. Only keying-free mean-centered Pearson combined near-best theory congruence (within .021 NMI), the best loading-shape match ($\bar{\phi} = .744$), and a genuine, well-conditioned correlation matrix (KMO = .971)—hence the main fit.

Item-Pair Agreement

Across all 1,225 item pairs, structure from language alone matched the raw correlations of 874,434 respondents at $r = .46$ (Figure 6).

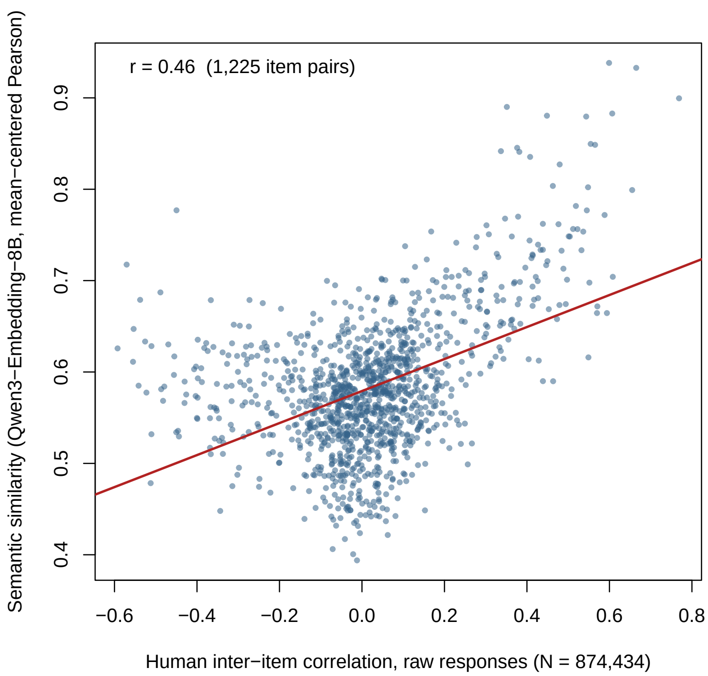

Figure 6

Semantic similarity (mean-centered Pearson encoding of the Qwen3-Embedding-8B item vectors) against human inter-item correlations from raw responses ($N = 874{,}434$), for all 1,225 item pairs; $r = .46$. The fitted regression line is shown in red.

Semantic–behavioral coherence is thus a measured relation: strong convergence for three domains, partial where warmth–sociability language blurs a boundary response covariance keeps sharp, and informative, encoding-specific divergence where the similarity model redefines "meaning the same thing."

Discussion

Summary of Findings

This article introduced `semanticfa`, an R package that runs the full exploratory factor-analytic workflow—similarity construction, retention, extraction and diagnostics, interpretation, and refinement—on scale items' semantic structure, response-free at every stage. The demonstration ran every exported function on the 50 IPIP Big-Five Factor Markers and validated results against a conventional factor analysis of $N = 874{,}434$ respondents, yielding three findings. First, the instrument's semantic structure is real and largely as intended: a multi-criterion consensus (parallel analysis 5, Kaiser 6, TEFI 2, EGA 5) retained five factors that recovered the Big Five domains, and an independent embedding-depth optimization agreed. Second, semantic–behavioral coherence is graded by domain: matched Tucker congruence with the human solution was .92 (Openness), .84 (Conscientiousness), .80 (Neuroticism), .64 (Extraversion), and .52 (Agreeableness), and item-pair similarities correlated $r = .46$ with raw-response correlations over all 1,225 pairs—useful convergence, informative divergence. Third, encoding item meaning is a substantive modeling decision: sign-flipped reverse-keyed embeddings became anti-topic vectors aligning *worst* with keyed human correlations ($r = .137$ vs. .434 unflipped), entailment- and cosine-based encodings had mirror-image strengths (Table 2), and Monte Carlo calibration showed conventional fit-index cutoffs do not transfer to embedding-derived matrices. Refinement proved most practical: eleven semantically redundant item pairs, a response-free 25-item short form that *exceeded* the full form in recovering the theoretical partition (NMI .527 to .810; ARI .402 to .571), and candidate-item vetting that caught all three planted flaws.

Theoretical Implications

The demonstration shows what estimating the Introduction's semantic–behavioral coherence buys. In the terms of Cronbach and Meehl (1955), the semantic and human analyses are qualitatively different operations—item-meaning geometry without respondents versus 874,434 people's response covariance—so their near-match for Openness, Conscientiousness, and Neuroticism ($\phi = .92$, .84, and .80) is intranetwork evidence that both engage the same network region. Under the realist caveat conceded at the outset, this strengthens the evidential webs without certifying that the attributes exist or cause anything (Borsboom et al., 2004): for these domains, the operationalized semantic proposal and response structure are two views of one organization.

The divergences—localized and interpretable, not diffuse—earn the relation its keep: the semantic solution splits Agreeableness and Extraversion into warmth/other-orientation and gregariousness factors rather than losing them, semantic Extraversion thereby matching human Agreeableness ($\phi = .52$) nearly as well as semantic Agreeableness did (Table 1)—a seam where English usage runs sociability and warmth together while response covariance holds them apart. The pattern has precedent: Kojima et al. (2026) found near-interchangeability at the general psychopathology factor (congruence = .97) but specific-factor divergences of non-semantic origin (insomnia fused with somatic complaints only in responses, plausibly physiological); Kmetty et al. (2021) found health-related occupations lower in embedding-derived prestige than in survey standing, plausibly via illness connotations; and Zhang and Qiu (2026) found the relation construct-dependent, weakest where baseline inter-item cohesion is strongest. Divergence thus signals its source: semantic compression of distinct behavioral dimensions, connotational drift, or behavioral covariance with causes wording does not carry.

Making coherence estimable also turns the Introduction's divergence conditions into testable predictions. If, as Borsboom (2008) argues, some clinical constructs are causal symptom networks, coherence should run systematically lower there than for common-cause constructs—an ordering `sfa_congruence()` makes falsifiable across instruments of both kinds; the logic extends to formative indicator sets (Bollen & Lennox, 1991) and method variance (Messick, 1995). An unexamined confound of every rating-scale factor analysis becomes a quantity theories can be required to predict.

Methodological Implications

The clearest lesson concerns reverse keying, the Introduction's fault line: a reverse-scored response keeps its content on an inverted scale; an embedding multiplied by -1 is an anti-topic vector. Hence `atomic_reversed` matched keyed human correlations worst of the cosine family ($r = .137$ vs. .434 unflipped), its checkerboard signs driving split-half reliability negative and rendering disattenuated congruence undefined—a pathology-diagnosing NA, reported, not suppressed. This substantiates the caution that embeddings do not differentiate reverse-keyed items (Suárez-Álvarez et al., 2026), explains why keyed sign-flipping fared worst in HEXACO pseudo-factor analyses (Guenole et al., 2025), fits the positive-sign bias surviving polarity fine-tuning (Hommel & Arslan, 2025), and settles negatively whether an embedding can be meaningfully "reversed" (Pellert et al., 2026). Content-not-key logic recurs package-wide: anchoring and projection both flagged N12 ("I am relaxed most of the time."), reverse-keyed yet semantically the inventory's most calm-pole statement (.82). Three upshots follow: prefer keying-free encodings (mean-centered Pearson, SQuID) or signed NLI over sign-flipping; build candidate-vetting reference fits without the scoring column, keeping centroids purely topical; and drop earlier short-form pipelines' drift-prone manual paraphrasing of reverse-keyed items (Jung & Seo, 2025; Wang et al., 2026), checking substitutions scale by scale.

Second, no single encoding or congruence metric dominates; each operationalizes a distinct inter-item relation—cosine symmetric and graded (Reimers & Gurevych, 2019), entailment directional over ordered premise–hypothesis pairs (Bowman et al., 2015). Profiles mirror accordingly: signed NLI best matched the theoretical partition (NMI $= .548$) and raw correlations ($r = .561$) but worst matched loading shape ($\bar{\phi} = .310$ vs. cosine's .744), while SQuID won keyed pair-level agreement ($r = .567$) on a near-unfactorable differenced matrix (KMO $= .022$). What language predicts is thus relation- and encoding-dependent, echoing item-parameter modeling: difficulty is text-predictable (Peters et al., 2025), discrimination largely not (Han et al., 2025), though simulation-based item-characteristic-curve reconstruction recovers it more readily (Ormerod, 2026). Two reporting practices follow: compare like with like—keying-free encodings to raw correlations, keyed to rescored; conflation under- or overstates coherence (NLI: $r_{\text{raw}} = .561$ vs. $r_{\text{keyed}} = .063$)—and report congruence on several metrics (loading shape, partition agreement, pair-level correlation), which can rank encodings oppositely.

Third, semantic-space fit needs matched chance baselines: by survey conventions the RMSR of .032 reads as good yet exceeded the entire structureless-embedding null range (.0119–.0134), the CAF (.412) fell below its nulls (.501–.532), and the TEFI (-27.71) beat all 100 null replicates (-21.66 to -17.82)—two of three conventional readings invert. This generalizes beyond confirmatory modeling the finding that chance meets conventional thresholds alarmingly often in embedding space (Pokropek, 2026) and answers the "no sample size" problem of Suárez-Álvarez et al. (2026): absent an N, Monte Carlo nulls supply empirical reference distributions; we propose such calibrated reporting as standard for semantic factor analyses. Retention repeats the pattern: semantic criteria disagree (parallel 5, Kaiser 6, TEFI 2, EGA 5) as simulations predict (Garrido et al., 2025; Golino, 2026), while human-side parallel analysis at $N = 874{,}434$ endorsed ten factors as trivially small minor factors cleared the noise threshold; such two-sided minor-factor proliferation is precisely the case for consensus-based retention with theory-fixed solutions (Goldberg, 1990).

Applications

The package's natural home is the stage Clark and Watson (2019) place first and weight most heavily—pre-data item-pool refinement—and its clinical analogues: screening large candidate pools, flagging redundancy early, reaching populations and settings where large validation samples are impractical (Kambeitz et al., 2025). The demonstration models that pass: redundancy screening exposed eleven within-domain near-paraphrase pairs (A24 $\sim$ A29 leading, wTO .462)—repetition inflating internal consistency and wording-driven correlations, not content (Christensen et al., 2023; Kojima et al., 2026); vetting correctly typed three new items as cross-loader (gap .01), near-duplicate (cosine .90), and weak match (.75 vs. domain average .81); and response-free shortening halved the instrument yet *improved* theoretical-partition recovery (NMI .527 to .810; ARI .402 to .571)—peripheral, not central, items had blurred construct boundaries (Huang et al., 2025; Kojima et al., 2026). The attenuation paradox (Clark & Watson, 2019) transfers: maximizing semantic homogeneity, like internal consistency, can prune breadth the construct requires, so an anchor-centrality-optimized short form still owes response-based validation—a caveat the package itself prints (Jung & Seo, 2025; Wang et al., 2026).

Second, response data may be scarce, absent, or compromised: Müller (2026) recovered semantic structure at $n = 55$ after the same instrument's confirmatory factor analysis collapsed, and `semanticfa`, needing no responses, lets structural screening precede recruitment. The synthetic respondents of Westwood (2025)—passing 99.8% of attention checks, biasing results toward inferred hypotheses—argue for analysis that touches no response pool and against model-derived structure passing as human evidence; hence the congruence machinery: semantic–behavioral coherence is demonstrated, never presumed—no current language model reproduces the human ability distribution (Liu et al., 2025), and virtual-respondent frameworks disclaim replicating human psychological processes (Lim et al., 2025).

Third, the tools scale across instruments: the jingle–jangle screen, rightly silent on the deliberately well-constructed Big-Five Factor Markers, targets cross-instrument screening, where label–content misalignment is endemic (Wulff & Mata, 2025, 2026), and `semanticfa`'s evaluation-time auditing of existing instruments complements AIGENIE's generation-time vetting of new item pools (Russell-Lasalandra et al., 2026). Finally, the design choices are methodological commitments: an explicit, programmatic embedding pipeline recovers structure that prompted chatbots do not (Huang et al., 2025); open, locally run models make sensitive-item analysis transparent and privacy-respecting (Hussain et al., 2024; Low et al., 2024); pinned model identity freezes the measurement instrument across reanalyses; and the pure-R, offline factor-analytic core (only the NLI matrix and live-embedding demonstrations need Python or the network) runs identically in a data-sovereignty-constrained clinic and a teaching laboratory. We commend the transcript-anchored reporting used here—every number checkable against a captured console transcript—to software papers.

Limitations

The central limitation is definitional: semantic factor analysis characterizes wording, not people; with no respondent, it asserts no causal path and supplies none of the response-process theory Borsboom et al. (2004) make the heart of validity (absent even from response-based factor structure), so coherence must never be reported as a validity coefficient. Of the six construct-validity aspects in Messick (1995), semantic evidence informs content (domain boundaries, representation, redundancy) and partly structural (scoring mirroring the semantic proposal); the substantive, generalizability, external, and consequential aspects require response and outcome data. Even the coherence thesis's strongest support bounds it: in Kojima et al. (2026), response-based factor scores predicted lifetime diagnoses better than text-based ones, textual information being effectively a subset of the behavioral. Hence the calibrated framings: a free, instant pilot study always followed by human data (Hommel & Arslan, 2025); no edge over aggregated expert judgment (Schoenegger et al., 2025); and, in Feraco and Toffalini (2025), no false promises of empirical fit though roughly half of misfit verdicts were false alarms—confidence better grounded than condemnation.

High coherence prices rather than rebuts the critique of Uher (2025), simultaneously grounding pre-data screening and measuring the share of "nomological" structure from ratings already in the items' inbuilt semantics—here $r = .46$ at the pair level, $\bar{\phi} = .744$ for factor shapes. Meaning-as-use also bounds the method: embeddings summarize a community's conventionalized regularities of use (Wittgenstein, 1953, §43), not the local construals raters form, so the idiosyncratic share of response variance is missed by design and perfect coherence is unexpected even from a flawless instrument.

Measurement is also model-dependent: corpus register, training window, and architecture change which human behaviors embedding similarities explain, none optimal across tasks (Mandera et al., 2017); the encoder is thus a declared instrument whose robustness merits reporting. The results rest on one model family (Qwen3-Embedding) and English items; even default full vectors can be suboptimal, signal concentrating in coordinate subsets (Golino, 2026)—witness the optimal depth of 1,431 of 4,096 here. Embedding methods trail human validators on behaviorally but not lexically phrased items (Milano, Ponticorvo, & Marocco, 2025), so behavioral instruments may recover less well than the trait-adjacent Big-Five Factor Markers; psychopathological nuance and hard-to-verbalize inner experience, likely underrepresented in training corpora, leave semantic structure least trustworthy where clinical assessment is hardest (Kambeitz et al., 2025).

Finally, the refinement tools carry caveats. Pruning presupposes reflective indicators—formative indicators are a census, not a sample, and removing one changes the construct (Bollen & Lennox, 1991)—so the measurement model must be settled first. The UVA cutoffs were calibrated on simulated response data, not semantic similarity matrices (Christensen et al., 2023). Semantically close items can still differ in response process (difficulty, extremity, framing)—similarity is not measurement equivalence (Wang et al., 2026)—or in theoretical identity, as with subjective stress versus the cortisol stress response; every purely semantic redundancy or jingle–jangle verdict—this package's included—should, as Wulff and Mata (2026) argue, inform expert deliberation, never replace it.

Future Directions

The most direct extension is comparative: coherence estimated across instruments, domains, and populations until it acquires norms. The demonstration—deliberately well-built, its coherence domain-graded—is one data point; the interesting variation lies between construct types: the symptom-network versus common-cause ordering derived above (Borsboom, 2008), construct-dependence in multilevel analyses (Zhang & Qiu, 2026), and the lexical-versus-behavioral phrasing gradient (Milano, Ponticorvo, & Marocco, 2025) await systematic test with a common toolkit. A second is methodological hardening: encoder robustness reported across embedding models as reliability is across raters (Mandera et al., 2017); embedding-depth selection promoted from diagnostic to tuning step, psychometric signal concentrating in coordinate subsets (Golino, 2026); redundancy cutoffs recalibrated on semantic similarity matrices, not inherited from response-data simulation (Christensen et al., 2023); and Monte Carlo calibration grown into community null baselines indexed by item count, embedding dimension, and encoding, so calibrated reporting (Pokropek, 2026) costs a lookup, not a simulation.

A third extends the pre-data use case across languages: structured generative adaptation performs at or near human benchmarks (Grobelny et al., 2025) but lacks a screen; the package suggests one—source and adapted items embedded in a shared multilingual space, congruence metrics repurposed as a pre-fielding check for structural drift in translation. All three keep the division of labor these results recommend. More than thirty years ago, Lara et al. (1992) predicted that semantic interpretation of factor solutions would have to be mechanized; embedding geometry has made it computable, and `semanticfa` makes it routine. What remains human—on this evidence, necessarily—is what wording cannot carry: response processes, criterion relations, and the judgment that turns a characterized proposal into a validated measure.

Conclusion

Every factor analysis of a rating scale rests on two structures: the covariance of responses and the organization of the language that elicited them. The first has a century of method, the second none until now; `semanticfa` brings the full discipline of exploratory factor analysis—explicit encoding choices, consensus retention, matched-null diagnostics, interpretation, and refinement—to that second structure, respondent-free and exactly reproducible from fixed embeddings. On a well-understood inventory it yielded a five-factor recovery of the Big Five; congruence with the 874,434-respondent structure strong for Openness, Conscientiousness, and Neuroticism, informatively weaker where English merges warmth and sociability; anti-topic vectors, not reverse-scored meanings, from sign-flipped reverse-keyed embeddings; and pre-data refinement that flagged redundant items, sharpened a short form, and vetted candidates. Agreement between the two structures—semantic–behavioral coherence—belongs in the nomological network of any construct measured in language: convergence shows words and respondents organizing the construct alike; divergence, where they do not and why. Neither replaces the other; semantic factor analysis estimates, before any respondent is recruited, how much of any scale's structure is already written in its items.

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Supplementary Materials

Full Loading Matrices

Tables S1 and S2 report the complete oblimin-rotated loading matrices for the five-factor semantic solution (mean-centered Pearson encoding of the Qwen3-Embedding-8B item vectors) and the five-factor human-response solution (minres/oblimin on keyed response correlations, $N = 874{,}434$), as written by the reproduction script to `output/semantic_loadings.csv` and `output/human_loadings.csv`.

Table S1

Semantic factor loadings for the 50 IPIP Big-Five Factor Marker items (mean-centered Pearson encoding, minres extraction, oblimin rotation)

Item	Domain	MR3	MR1	MR2	MR5	MR4
E1	Extraversion	.093	.308	.076	.354	.148
E2	Extraversion	.051	.196	.449	.203	.122
E3	Extraversion	.084	.061	.127	.752	.013
E4	Extraversion	−.001	.546	.206	.161	.008
E5	Extraversion	.037	.205	.117	.385	.257
E6	Extraversion	.093	.175	.432	.065	.190
E7	Extraversion	.005	.110	.120	.509	.182
E8	Extraversion	.062	.220	.570	.169	−.044
E9	Extraversion	−.001	.232	.225	.450	.066
E10	Extraversion	.094	.034	.408	.390	.014
N11	Neuroticism	.817	.073	−.001	.013	.023
N12	Neuroticism	.176	.395	.079	.347	−.093
N13	Neuroticism	.490	.177	.154	.022	.157
N14	Neuroticism	.297	.170	.319	.141	.022
N15	Neuroticism	.801	.074	.069	.054	−.042
N16	Neuroticism	1.003	−.020	.010	−.053	−.029
N17	Neuroticism	.790	.034	−.009	.058	.053
N18	Neuroticism	.860	.027	−.087	.019	.029
N19	Neuroticism	.871	.057	.055	−.071	−.012
N20	Neuroticism	.574	.086	.104	.096	.082
A21	Agreeableness	.149	.039	.718	.128	−.043
A22	Agreeableness	.041	.030	.230	.485	.261
A23	Agreeableness	.267	.193	.146	.211	.113
A24	Agreeableness	.453	−.082	.069	.425	.141
A25	Agreeableness	.084	.000	.786	.025	−.031
A26	Agreeableness	.390	.106	.043	.303	.100
A27	Agreeableness	−.023	.030	.895	.072	−.029
A28	Agreeableness	.077	.459	.171	.308	−.026
A29	Agreeableness	.553	−.158	.073	.390	.119
A30	Agreeableness	.179	.203	.046	.596	−.001
C31	Conscientiousness	.050	.626	−.056	.159	.139
C32	Conscientiousness	.091	.617	.144	.070	−.029

Table S1 (continued)

Item	Domain	MR3	MR1	MR2	MR5	MR4
C33	Conscientiousness	.064	.566	.041	.091	.233
C34	Conscientiousness	.354	.503	.081	−.046	.070
C35	Conscientiousness	.103	.716	.004	−.001	.050
C36	Conscientiousness	.207	.566	.125	−.103	.022
C37	Conscientiousness	.082	.567	.048	.121	.103
C38	Conscientiousness	.085	.567	.336	−.094	−.048
C39	Conscientiousness	.082	.677	−.034	.093	.066
C40	Conscientiousness	.074	.605	.046	−.032	.207
O41	Openness	.063	.263	−.034	.222	.450
O42	Openness	.244	−.122	.350	−.095	.563
O43	Openness	.181	.155	−.027	.169	.554
O44	Openness	−.045	.071	.615	−.128	.403
O45	Openness	.012	.300	−.036	.126	.582
O46	Openness	.155	−.003	.438	−.093	.437
O47	Openness	.220	.205	−.009	.217	.332
O48	Openness	.162	.235	.070	.086	.425
O49	Openness	.164	.351	.100	.098	.250
O50	Openness	.069	.249	−.024	.150	.606

Table S2

Human-response factor loadings for the same 50 items (keyed correlations, $N = 874{,}434$; minres extraction, oblimin rotation)

Item	Domain	MR4	MR1	MR3	MR2	MR5
E1	Extraversion	.712	.042	−.010	−.015	−.012
E2	Extraversion	.713	.086	.053	−.026	.008
E3	Extraversion	.623	−.189	.179	.061	−.068
E4	Extraversion	.755	−.023	−.039	.018	−.027
E5	Extraversion	.714	.012	.123	.067	.034
E6	Extraversion	.522	.005	.073	.010	.235
E7	Extraversion	.721	−.004	.069	.012	−.019
E8	Extraversion	.618	.039	−.109	−.070	.026
E9	Extraversion	.653	.027	−.105	−.050	.103
E10	Extraversion	.684	−.079	−.018	.014	−.019
N11	Neuroticism	−.074	.708	.125	.061	−.055
N12	Neuroticism	−.064	.559	.013	.101	−.004
N13	Neuroticism	−.125	.611	.208	.092	.017
N14	Neuroticism	−.129	.349	.059	−.069	.092
N15	Neuroticism	.005	.505	.004	−.029	−.095
N16	Neuroticism	.027	.746	.031	.005	−.068
N17	Neuroticism	.077	.719	−.021	−.091	.008
N18	Neuroticism	.063	.741	−.035	−.096	.005
N19	Neuroticism	.055	.719	−.175	.041	−.017
N20	Neuroticism	−.219	.574	.017	−.135	.114
A21	Agreeableness	−.049	−.056	.499	−.006	.061
A22	Agreeableness	.282	−.041	.522	−.040	.060
A23	Agreeableness	−.180	−.240	.418	.144	−.071
A24	Agreeableness	−.055	.033	.807	−.007	−.004

Table S2 (continued)

Item	Domain	MR4	MR1	MR3	MR2	MR5
A25	Agreeableness	.060	−.027	.661	−.038	−.002
A26	Agreeableness	−.063	.131	.618	.001	−.074
A27	Agreeableness	.225	−.084	.613	−.035	.015
A28	Agreeableness	.077	−.014	.559	.061	.001
A29	Agreeableness	.019	.100	.700	.019	.033
A30	Agreeableness	.260	−.100	.362	.086	.062
C31	Conscientiousness	.015	.000	−.032	.648	.091
C32	Conscientiousness	−.051	−.035	−.069	.573	−.105
C33	Conscientiousness	−.068	.075	.062	.405	.251
C34	Conscientiousness	.012	−.279	−.010	.556	−.003
C35	Conscientiousness	.073	.009	−.002	.632	−.079
C36	Conscientiousness	−.024	−.088	−.037	.614	−.043
C37	Conscientiousness	−.049	.160	−.004	.581	.040
C38	Conscientiousness	.006	−.152	.097	.465	.049
C39	Conscientiousness	.053	.111	.058	.634	−.054
C40	Conscientiousness	.002	.052	.009	.456	.249
O41	Openness	−.015	−.011	−.058	.035	.596
O42	Openness	−.068	−.189	.014	−.030	.579
O43	Openness	−.014	.107	.091	−.099	.545
O44	Openness	−.095	−.109	.114	−.100	.518
O45	Openness	.166	−.027	−.069	.122	.575
O46	Openness	.011	−.032	.091	−.052	.499
O47	Openness	.019	−.115	−.044	.154	.474
O48	Openness	−.006	.076	−.126	−.024	.564
O49	Openness	−.185	.115	.187	.037	.396
O50	Openness	.122	.017	.008	.000	.662